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1  -- =====
2  -- SECTION B : Filtering Queries
3  -- Database : Celebal_Week2
4  -- Table : dbo.Superstore
5  -- Tool : SQL Server Management Studio (SSMS)
6  -- Author : ARATI THORAT | Celebal Internship - Week 2
7  -- =====
8
9  USE Celebal_Week2;
10 GO
11
12 -- -----
13 -- B1. Filter by Region
14 -- PURPOSE : Isolate sales from a specific geography
15 -- -----
16
17 -- All orders from the West region
18 SELECT
19     Order_ID, Customer_Name, City, State,
20     Sales, Profit
21 FROM     dbo.Superstore
22 WHERE    Region = 'West'
23 ORDER BY Sales DESC;
24
25 -- Orders from East OR Central
26 SELECT
27     Order_ID, Customer_Name, Region,
28     Sales, Profit
29 FROM     dbo.Superstore
30 WHERE    Region IN ('East', 'Central')
31 ORDER BY Region, Sales DESC;
32
33 /*
34     OBSERVATION:
35     West region has the highest number of orders among all regions.
36     California dominates the West with the most transactions.
37 */
38
39
40 -- -----
41 -- B2. Filter by Product Category
42 -- PURPOSE : Drill into category-specific performance
43 -- -----
44
45 -- Technology products only
46 SELECT
47     Product_ID, Product_Name, Sub_Category,
48     Sales, Profit, Quantity
49 FROM     dbo.Superstore
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50 WHERE Category = 'Technology'
51 ORDER BY Sales DESC;
52
53 -- Office Supplies in a specific sub-category
54 SELECT
55     Product_Name, Sub_Category,
56     Sales, Profit, Discount
57 FROM   dbo.Superstore
58 WHERE  Category      = 'Office Supplies'
59 AND    Sub_Category = 'Binders'
60 ORDER BY Sales DESC;
61
62 /*
63     OBSERVATION:
64     Technology has the highest individual sales values (Phones, Copiers).
65     Binders are the most ordered sub-category by volume.
66 */
67
68
69 -- _____
70 -- B3. Filter by Date Range
71 --     PURPOSE : Analyse sales within specific time windows
72 -- _____
73
74 -- Orders placed in the year 2017
75 SELECT
76     Order_ID, Order_Date, Ship_Date,
77     Customer_Name, Sales, Profit
78 FROM   dbo.Superstore
79 WHERE  Order_Date BETWEEN '2017-01-01' AND '2017-12-31'
80 ORDER BY Order_Date;
81
82 -- Q1 2017 orders (January to March)
83 SELECT
84     Order_ID, Order_Date,
85     Customer_Name, Region, Sales
86 FROM   dbo.Superstore
87 WHERE  Order_Date BETWEEN '2017-01-01' AND '2017-03-31'
88 ORDER BY Order_Date;
89
90 -- Orders shipped using Standard Class in 2016
91 SELECT
92     Order_ID, Order_Date, Ship_Date,
93     Ship_Mode, Customer_Name, Sales
94 FROM   dbo.Superstore
95 WHERE  Ship_Mode  = 'Standard Class'
96 AND    Order_Date BETWEEN '2016-01-01' AND '2016-12-31'
97 ORDER BY Order_Date;
98
```

```
99  /*
100  OBSERVATION:
101  The dataset spans 2014-2017. Year-over-year queries show
102  steady sales growth. Q4 (Oct-Dec) consistently shows
103  the highest sales volume across all years.
104  */
105
106
107  -- -----
108  -- B4. Filter by Sales Amount
109  -- PURPOSE : Identify high-value and low-value transactions
110  -- -----
111
112  -- High-value orders (Sales > $1,000)
113  SELECT
114      Order_ID, Customer_Name, Product_Name,
115      Category, Sales, Profit
116  FROM    dbo.Superstore
117  WHERE   Sales > 1000
118  ORDER BY Sales DESC;
119
120  -- Orders with Sales between $500 and $1,000
121  SELECT
122      Order_ID, Customer_Name, Sales, Profit
123  FROM    dbo.Superstore
124  WHERE   Sales BETWEEN 500 AND 1000
125  ORDER BY Sales DESC;
126
127  -- Very small orders (Sales < $10)
128  SELECT TOP 20
129      Order_ID, Customer_Name, Product_Name,
130      Sales, Quantity, Discount
131  FROM    dbo.Superstore
132  WHERE   Sales < 10
133  ORDER BY Sales ASC;
134
135  /*
136  OBSERVATION:
137  Orders above $1,000 are dominated by Technology (Copiers, Phones).
138  Very small orders (< $10) often have high discounts applied,
139  which erodes profit significantly.
140  */
141
142
143  -- -----
144  -- B5. Filter by Discount applied
145  -- PURPOSE : Understand impact of discounting strategy
146  -- -----
147
```

```
148 -- Items with NO discount
149 SELECT
150     Order_ID, Product_Name, Category,
151     Sales, Discount, Profit
152 FROM   dbo.Superstore
153 WHERE  Discount = 0
154 ORDER BY Sales DESC;
155
156 -- Heavily discounted items (Discount > 30%)
157 SELECT
158     Order_ID, Product_Name, Category,
159     Sales, Discount, Profit
160 FROM   dbo.Superstore
161 WHERE  Discount > 0.30
162 ORDER BY Discount DESC;
163
164 /*
165     OBSERVATION:
166     Items with 0% discount are generally more profitable.
167     Items with discount > 30% often show NEGATIVE profit -
168     indicating over-discounting is hurting the business.
169 */
170
171
172 -- -----
173 -- B6. Filter Loss-Making Orders (Profit < 0)
174 --     PURPOSE : Identify orders that lost money
175 -- -----
176 SELECT
177     Order_ID, Customer_Name, Product_Name,
178     Category, Sub_Category,
179     Sales, Discount, Profit
180 FROM   dbo.Superstore
181 WHERE  Profit < 0
182 ORDER BY Profit ASC;
183
184 /*
185     OBSERVATION:
186     There are ~1,871 loss-making line items (about 18.7% of all rows).
187     Most losses are in Furniture (Tables) and Office Supplies
188     categories where discounts exceed 40%.
189 */
190
191
192 -- -----
193 -- B7. Filter using LIKE (pattern matching)
194 --     PURPOSE : Search for products/customers by partial name
195 -- -----
196
```

```
197 -- Products containing the word 'Table'
198 SELECT
199     Product_ID, Product_Name, Category, Sub_Category,
200     Sales, Profit
201 FROM   dbo.Superstore
202 WHERE  Product_Name LIKE '%Table%';
203
204 -- Customers whose name starts with 'A'
205 SELECT DISTINCT
206     Customer_ID, Customer_Name, Segment, Region
207 FROM   dbo.Superstore
208 WHERE  Customer_Name LIKE 'A%'
209 ORDER BY Customer_Name;
210
211 -- Orders starting with CA- (California convention)
212 SELECT TOP 10
213     Order_ID, Order_Date, Customer_Name, State
214 FROM   dbo.Superstore
215 WHERE  Order_ID LIKE 'CA-%'
216 ORDER BY Order_Date DESC;
217
218 /*
219     OBSERVATION:
220     LIKE is powerful for exploring semi-structured data.
221     'Table' products (Furniture) consistently show negative profit -
222     a critical business finding.
223 */
224
225
226 -- _____
227 -- B8. Filter by Segment
228 --     PURPOSE : Compare performance across customer types
229 -- _____
230
231 -- Corporate segment customers only
232 SELECT DISTINCT
233     Customer_ID, Customer_Name, City, State, Region
234 FROM   dbo.Superstore
235 WHERE  Segment = 'Corporate'
236 ORDER BY Customer_Name;
237
238 -- Home Office customers in the South region
239 SELECT DISTINCT
240     Customer_ID, Customer_Name, City, State
241 FROM   dbo.Superstore
242 WHERE  Segment = 'Home Office'
243 AND    Region = 'South'
244 ORDER BY State;
245
```

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246  /*
247      OBSERVATION:
248      Corporate segment places fewer but larger individual orders.
249      Home Office customers are concentrated in the South and West.
250  */
251
252
253  -- -----
254  -- B9. Filter using NOT and NOT IN
255  --     PURPOSE : Exclude specific groups from analysis
256  -- -----
257
258  -- Orders NOT shipped via Standard Class
259  SELECT
260      Order_ID, Order_Date, Ship_Mode,
261      Customer_Name, Sales
262  FROM    dbo.Superstore
263  WHERE   Ship_Mode <> 'Standard Class'
264  ORDER BY Ship_Mode, Sales DESC;
265
266  -- Products NOT in Furniture category
267  SELECT
268      Product_ID, Product_Name, Category
269  FROM    dbo.Superstore
270  WHERE   Category NOT IN ('Furniture')
271  ORDER BY Category, Product_Name;
272
273  /*
274      OBSERVATION:
275      Non-Standard Class shipping (First Class, Same Day) represents
276      premium shipping and is used more often by Corporate customers.
277  */
278
279
280  -- -----
281  -- B10. Combined AND / OR filter
282  --     PURPOSE : Apply multiple business conditions simultaneously
283  -- -----
284
285  -- High-sales OR high-discount transactions
286  SELECT
287      Order_ID, Customer_Name, Category,
288      Sales, Discount, Profit
289  FROM    dbo.Superstore
290  WHERE   Sales > 500
291  OR      Discount > 0.30
292  ORDER BY Sales DESC;
293
294  -- High-value AND profitable orders
```

```
295 SELECT
296     Order_ID, Customer_Name,
297     Product_Name, Sales, Profit
298 FROM   dbo.Superstore
299 WHERE  Sales > 500
300 AND    Profit > 100
301 ORDER BY Profit DESC;
302
303 -- Corporate segment AND Technology category
304 SELECT
305     Order_ID, Customer_Name,
306     Product_Name, Sales, Profit
307 FROM   dbo.Superstore
308 WHERE  Segment = 'Corporate'
309 AND    Category = 'Technology'
310 ORDER BY Sales DESC;
311
312 /*
313     OBSERVATION:
314     Corporate + Technology is the most profitable combination -
315     high sales, low discount, strong profit margin.
316     High discount + high sales → often still ends in loss.
317 */
318
319
320 -- -----
321 -- B11. TOP N results (Limit output)
322 --     PURPOSE : Get best/worst performers quickly
323 -- -----
324
325 -- Top 10 highest sales transactions
326 SELECT TOP 10
327     Order_ID, Customer_Name, Product_Name,
328     Category, Sales, Profit
329 FROM   dbo.Superstore
330 ORDER BY Sales DESC;
331
332 -- Top 10 worst profit (biggest losses)
333 SELECT TOP 10
334     Order_ID, Customer_Name, Product_Name,
335     Category, Sales, Discount, Profit
336 FROM   dbo.Superstore
337 ORDER BY Profit ASC;
338
339 -- Top 10 most discounted items
340 SELECT TOP 10
341     Order_ID, Product_Name, Category,
342     Sales, Discount, Profit
343 FROM   dbo.Superstore
```

```
344 ORDER BY Discount DESC;
345
346 /*
347     OBSERVATION:
348     Top 10 highest sales are all Technology (Copiers, Phones, Machines).
349     Top 10 worst losses include Tables and Bookcases with 40-80% discounts.
350     Heavy discounting is the #1 driver of losses in this dataset.
351 */
```